

Johnson C. Smith University
Introductory Abstract Algebra I | MTH 335 A | Spring 2013

Pure mathematics is, in its way, the poetry of logical ideas.
~Albert Einstein

A problem worthy of attack, proves its worth by fighting back!
~Paul Erdos

Instructor Information

Name: Dr. James Ashe

Office location: 307 Davis Hall (back); office hours will also be held in the Math Lab, EDU 115

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Office hours: MWF 10–11am and 12–1pm; By appointment only, MW 8–9am and 2–3pm.

Course Description

This is a **3 credit hour** course; the first in a two-course sequence which are offered for students from the sciences, social sciences and engineering. Topics covered: Elementary set theory and logic. Mappings, groups, rings, integral domains, fields, and polynomials. Prerequisite: Mathematics 231 or consent of Department.

Instructional methods will consist of lectures, in-class discussions, exams, and student presentations. Other methods may be adopted according to the particular needs of the class.

Course Objectives

The main objective of this course is to develop mathematical reasoning by learning how to read, write, and present proofs. Students completing this course will demonstrate knowledge of the fundamental concepts in the core areas listed in the above section. Additionally, students should demonstrate the ability for abstract reasoning, rigor, and logical thinking.

Course Materials

Suggested Textbook: There is no required textbook. Below is a list of recommended reference books. The first three are free.

Abstract Algebra: Theory and Applications by Thomas W. Judson

A Gentle Introduction to Abstract Algebra B. A Sethuraman

Abstract Algebra: The Basic Graduate Year by Robert Ashe

Contemporary Abstract Algebra by Joseph A. Gallian

Calculator: Feel free to bring one if you want.

Grades

Grading scale: $A \geq 90 > B \geq 80 > C \geq 70 > D \geq 60 > F$

Evaluation: Grades will be taken from homework, quizzes, and Exams as follows:

Homework= 50%; Presentations= 15%; Midterm= 15%; Final = 20%

Homework: Regular homework assignments will be given. Students can collaborate on homework assignments, however each student should hand in their own assignments. The lowest homework score will be dropped.

Presentations: Each student will be given one problem to present to the class from each homework assignment. The date of presentation will typically precede the homework due date. You will be assessed on

both your ability to work the problem and explain it.

Midterm and Final Exam: Both the midterm and final will be cumulative exams. The midterm will be given on Friday, March 22, and the final will be given in our normal class room on Friday, May 17 from 1-3pm; see the exam schedule:

www.jcsu.edu/happenings/exam-schedule/spring-exam. *If beneficial, the final exam score will replace the midterm exam score.*

Class Policies

JCSU policies regarding attendance, academic integrity, grades, and change of enrollment will be enforced. It is the student's responsibility to be aware of these policies and of related deadlines as well as any announcements or syllabus adjustments (written or oral) made in class. Please see the student handbook for details, jcsu.edu/docs/handbook.pdf.

Attendance: Class attendance is required for all JCSU students. Each student is allowed as many hours of absence per term as credit hours(s) received (not to exceed 4) for the class. Students who exceed the maximum number of absences may receive a failing grade for the course.

Students who may miss classes while representing the University in an official capacity are exempt from regulations governing absences. However, absence from class for official University business does not relieve the student from responsibility for any class assignments that may be missed during the period of absence. In such cases, *it is the student's responsibility to make arrangements to make up any work that will be missed before the assignment/quiz/exam is missed.*

Late Homework: An assignment will have 25% deducted for each day it is late.

Missed presentations: A student will be allowed one make-up for a missed presentation. Presentations can be rescheduled if a University sponsored event conflicts with the presentation date, and advance notice of the absence was given by the student.

Missed exams: Missed exams may be made up at the discretion of the Instructor. (1) you must notify me by email or phone within 24 hours of the missed exam AND (2) you must have no more than 3 unexcused absences in the class. If BOTH of these conditions are met, a make-up will be given. However unless a documented excuse is provided, 10% will automatically be deducted from the make-up. If you will be gone for any reason, it will be *your* responsibility to inform me at least a week in advance to schedule to take the exam early.

Other

Important Dates

Friday 22 nd March, 2013	Exam 4
Friday 17 th May, 2013	Final Exam

Student support services: If you need course adaptations or accommodations for a documented disability please contact the Office of Disability Services.

jcsu.edu/directory/federal-requirements/general-institutional-information/jcsu-disability-services,
phone: (704) 398-1116.

The Office of Counseling Services provides confidential intakes and assessments, personal counseling and referrals to appropriate resources to all enrolled students on a walk-in or appointment basis.

jcsu.edu/admissions/future_students/meet_your_counselor, phone (704) 378-1044.

Disclaimer: While I do not foresee any significant changes to the items in this syllabus during the semester, I reserve the right to make appropriate changes. Students will be notified of any such changes in writing.

JCSU's Fall 2013 academic calendar can be found here: jcsu.edu/happenings/academic-calendar

The mathematician does not study pure mathematics because it is useful; he studies it because he delights in it and he delights in it because it is beautiful.

~Poincare